

STA201 Assignment 5

1. A rental car company claims the mean time to rent a car on their website is 60 seconds with a standard deviation of 22 seconds. A random sample 36 customers attempted to rent a car on the website. The mean time to rent was 55 seconds. Is this enough evidence to say that the mean time is less than the company's claim? Consider $\alpha=0.05$.
2. A teacher wanted to know how well the gifted students in her class perform relative to her other classes. She administers a standardized test with a mean of 50 and standard deviation of 10. Her class of 31 students has an average score of 55. Test the claim that their average score is higher than 50? Consider $\alpha=0.05$.
3. A researcher wanted to study the effects of mentoring on intelligence scores. He wanted to know as a baseline what the average intelligence of his students were relative to the general population. He used a standardized IQ test which has a mean of 100 and standard deviation of 16. The 50 students in his study scored an average of 102 on the IQ test. Does his study support the baseline? Consider $\alpha=0.05$.
4. Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that mean burning rate must be 50 centimeters per second. The experimenter selects a random sample of 25 propellants and obtains a sample average burning rate of 48.7 centimeters per second with standard deviation of 2 centimeters per second. What conclusion can be drawn from the samples?
5. You have just taken ownership of a pizza shop. The previous owner told you that you would save money if you bought the mozzarella cheese in a 4.5 pound slab. Each time you purchase a slab of cheese, you weigh it to ensure that you are receiving 72 ounces of cheese. The results of 7 random measurements are 70, 69, 73, 68, 71, 69 and 71 ounces. Are these differences due to chance or is the distributor giving you less cheese than you deserve?